Basin Shaping for Opponent-Aware Learning: Local Meta-MAPG Convergence with Restarts

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Abstract

Multiagent reinforcement learning is difficult because each agent experiences the others as a moving target. Meta-MAPG addresses this non-stationarity by differentiating through both an agent's own future updates and the future updates of its peers, but its local stochastic-approximation status has been unclear. We study Meta-MAPG from a local convergence viewpoint. Our analysis treats the finite-unroll Meta-MAPG correction as an opponent-aware shaping field added to the base game-gradient dynamics. In phase 1, constant shaping is used only as a basin-entry mechanism: if the shaped field improves local drift by $\lambda\mu_M$, the certified entry time shrinks by the same factor. In phase 2, the shaping coefficient is annealed to zero, so the limiting ODE returns to the original game-gradient flow and Giannou et al.'s stable-Nash theorem applies. Independent restarts then globalise the local guarantee with an almost-sure finite-restart bound and a geometric tail. Experiments in Stag Hunt and iterated prisoner's dilemma match this picture: peer-learning terms drive the empirical gain, restart budget becomes equilibrium-selection budget, basin maps expand under opponent-aware shaping, and a small MLP IPD check shows the effect is not purely tabular.

1. Introduction

Learning in stochastic games is hard because every learner changes the environment faced by the others. From the perspective of any single agent, the environment is therefore effectively non-stationary. Giannou et al. (2022) prove local convergence of policy gradients to stable Nash policies under strategic stability and second-order sufficiency. This is the right baseline for policy-gradient dynamics near a stable Nash point, but it still treats opponent learning as background variation rather than something the update itself can model.

Meta-learning suggests a richer view. Kim et al. (2021) model multiagent learning as a Markov chain of joint policy updates and derive Meta-MAPG, whose meta-gradient contains a current-policy term, an own-learning term, and a peer-learning term. This decomposition captures the fact that an agent's current behaviour changes both its own future policy and the future policies of the peers it is interacting with. Empirically, those extra terms matter. The open question is when they fit the local stochastic-approximation picture used by the standard convergence theory.

Our paper takes that local question as primary. We treat the finite-unroll Meta-MAPG correction as an opponent-aware shaping field added to the base game-gradient dynamics. The central idea is to separate two roles that are often conflated. Constant shaping can be useful for finite-time basin entry and equilibrium selection, but asymptotic convergence should still target the stable Nash equilibria of the original game. We therefore analyze a two-phase schedule: a constant-shaping phase for basin entry, followed by an annealed phase that turns the limiting ODE back into the original game-gradient flow.

Our contributions.

1) **Meta-MAPG as local stochastic approximation.** Under bounded rewards, smooth policies, controlled unroll length, and standard sampling assumptions, we show that the finite-unroll Meta-MAPG update can be written as the base game-gradient field plus controlled opponent-aware terms with martingale noise and summable bias.
2) **A two-phase local convergence picture.** We show that constant opponent-aware shaping can improve local drift and certified basin geometry around a desirable equilibrium, while an annealed second phase restores the original game's stable Nash target.
3) **Restart-based globalisation.** We turn the local theo-

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rem into a global search statement: with positive restart mass on the basin, independent restarts yield an almost-sure finite-restart guarantee and a geometric tail bound. Restart budget becomes equilibrium-selection budget.

4) **Theory-facing experiments.** Stag Hunt and iterated prisoner's dilemma test the mechanism predicted by the theory: peer-learning changes equilibrium selection, basin maps expand under shaping, and the two-phase schedule preserves the intended target while improving basin entry.

2. From Stochastic Games to Meta-MAPG

2.1. Stochastic Games

We consider finite discounted stochastic games

$$\mathcal{G} = (N, \mathcal{S}, \{\mathcal{A}_i\}_{i\in N}, P, \{r_i\}_{i\in N}, \rho, \gamma_{\text{disc}}),$$

with finite state space $\mathcal{S}$, finite action sets $\mathcal{A}_i$, transition kernel P, bounded rewards $|r_i| \le R_{\max}$, initial-state distribution ρ, and discount factor $\gamma_{\text{disc}} \in [0, 1)$. Each agent has a differentiable stationary Markov policy $\pi_{\phi_i}(a_i \mid s)$ with parameter $\phi_i \in \mathbb{R}^{d_i}$; the joint parameter is $\phi = (\phi_i)_{i=1}^N \in \mathbb{R}^d$. The value of agent i is

$$J_i(\phi) = \mathbb{E}_{\rho,\pi_\phi}\left[\sum_{t\ge 0} \gamma_{\text{disc}}^t r_i(s_t, a_t)\right],$$

and the standard game-gradient field is

$$v(\phi) = \left(\nabla_{\phi_i} J_i(\phi)\right)_{i=1}^N.$$

First-order stationary points of this field correspond to Nash policies under the gradient-dominance property used by Giannou et al. (2022); finite stochastic games possess stationary Nash equilibria (Fink, 1964).

Definition 2.1 (Stable Nash and SOS)**.** A Nash parameter ϕ^* is stable if

$$\langle v(\phi), \phi - \phi^* \rangle < 0$$

for all $\phi \ne \phi^*$ sufficiently close to ϕ^*. It is second-order stationary (SOS) if there exist $\mu > 0$ and $\rho > 0$ such that

$$\langle v(\phi), \phi - \phi^* \rangle \le -\mu \|\phi - \phi^*\|^2$$

whenever $\|\phi - \phi^*\| \le \rho$.

The SOS condition is the quantitative stability condition that yields an attracting neighbourhood and a rate in Giannou et al. (2022). We use it as the local anchor.

2.2. A Markov Chain of Joint Policies

Following Kim et al. (2021), we view learning as a Markov chain of joint policy parameters $\phi_0, \dots, \phi_L$, where each inner transition is produced by a policy-gradient update. For agent i, Meta-MAPG decomposes the meta-gradient into

$$G_i^{\text{MAPG}} = G_i^{\text{cur}} + G_i^{\text{own}} + G_i^{\text{peer}}.$$

The current term is the usual policy-gradient score term. The own-learning term differentiates through how ϕ_i^0 affects agent i's adapted parameters $\phi_i^{1:L}$. The peer-learning term differentiates through how ϕ_i^0 affects other agents' future parameters $\phi_{-i}^{1:L}$. The peer term is the Meta-MAPG analogue of LOLA-style opponent shaping; the own term is absent from pure LOLA.

We write the population opponent-aware field as

$$F_L(\phi) = v(\phi) + M_L^{\text{own}}(\phi) + M_L^{\text{peer}}(\phi),$$

where L is the unroll length. In the convergence theorem we use the annealed update

$$\phi_{n+1} = \text{Proj}_{\mathcal{K}}\left[\phi_n + \alpha_n\left(\hat{v}_n + \lambda_n \hat{M}_{L_n,n}\right)\right], \quad (1)$$

Here $\mathcal{K} \subset \mathbb{R}^d$ is compact, $\hat{M}_{L_n,n}$ estimates the combined own/peer correction, and λ_n controls the asymptotic shaping strength. Projection can be replaced by any standard boundedness mechanism. We state it explicitly so the projection lemma of Kushner & Clark (1978) and the ODE tracking argument of Borkar (2008) apply without extra tightness work.

Definition 2.2 (Nash-consistent shaping)**.** Let U be a neighbourhood of a stable Nash set $\mathcal{N}$ of the base game-gradient field v. A scheduled opponent-aware correction $\lambda_n M_{L_n}$ is Nash-consistent on U if its contribution to the stochastic-approximation recursion is either summable,

$$\sup_{\phi\in U} \|\lambda_n M_{L_n}(\phi)\| \le \beta_n, \qquad \sum_n \alpha_n \beta_n < \infty,$$

so the limiting ODE on U is $\dot{\phi} = v(\phi)$, or else its non-vanishing limit field $v + M$ has exactly the same isolated SOS equilibria in U as v with the same local attracting neighbourhoods. The convergence theorem uses the first, annealed case; the second case only records when constant shaping is locally equilibrium-preserving.

3. Local Assumptions

The assumptions separate standard stochastic approximation from the Meta-MAPG estimator step.

Assumption 1 (Step sizes and bounded iterates)**.** The step sizes satisfy $\alpha_n > 0$, $\sum_n \alpha_n = \infty$, and $\sum_n \alpha_n^2 < \infty$. Iterates remain in a compact set $\mathcal{K}$, either by projection in (1) or by an equivalent stability mechanism.

Assumption 2 (Estimator decomposition)**.** There is a locally Lipschitz effective field F—equal to v in the annealed

case and to an equilibrium-preserving limit $v + M$ in the non-vanishing case (Theorem 2.2)—such that the sampled update $g_n = \hat{v}_n + \lambda_n \hat{M}_{L_n,n}$ admits the decomposition

$$g_n = F(\phi_n) + \xi_{n+1} + b_n,$$

where $\{\xi_{n+1}\}$ is a martingale-difference sequence adapted to $\mathcal{F}_n$ with

$$\mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] = 0, \qquad \mathbb{E}[\|\xi_{n+1}\|^2 \mid \mathcal{F}_n] \le \sigma^2,$$

and $\|b_n\| \le C\beta_n$ for a deterministic sequence $\beta_n \ge 0$.

Assumption 3 (Truncation and modeling bias)**.** The unroll schedule L_n and opponent-modeling procedure produce a residual finite-unroll/modeling error absorbed into the bias sequence β_n of Assumption 2.

(a) *Phase 1 (bounded).* Over any finite constant-shaping window, $\beta_n \le \beta_0$ is uniformly bounded.

(b) *Phase 2 (vanishing).* Under the annealed schedule of Theorem 4.4, the unroll length grows so that $\beta_n \to 0$ and $\sum_n \alpha_n \beta_n < \infty$. A sufficient choice is $L_n = \lceil r \log n / \log(1/\gamma_{\text{disc}}) \rceil$ for any $r > 1 - p$, giving $\beta_n = O(n^{-r})$.

Assumption 4 (Summability and annealing)**.** Step sizes, bias, and shaping coefficients jointly satisfy

$$\sum_n \alpha_n \beta_n < \infty, \qquad \sum_n \alpha_n \lambda_n < \infty,$$

with M_{L_n} uniformly bounded on $\mathcal{K}$. Under these conditions the limiting ODE of (1) is the base game-gradient flow $\dot{\phi} = v(\phi)$, so the opponent-aware correction is asymptotically transient. Phase 1 of Theorem 4.4 relaxes the $\sum_n \alpha_n \lambda_n < \infty$ requirement to a constant $\lambda_n = \lambda_c$ over a finite-time window only; phase 2 restores it.

Assumption 5 (Strategic stability)**.** The stable set $\mathcal{N}$ consists of isolated SOS Nash equilibria satisfying Theorem 2.1. In the non-annealed local geometry statement, the shaped field $F = v + \lambda M$ satisfies the analogous SOS condition around the same equilibrium.

Remark 3.1. The convergence target is the original game's stable Nash set $\mathcal{N}$. If shaping is kept constant, meta-objectives can have non-Nash fixed points. The theorem is therefore annealed, or explicitly equilibrium-preserving. Constant shaping is used for finite-time basin geometry in Theorem 4.3, not to claim that arbitrary meta-game stationary points are Nash.

4. Local Meta-MAPG with Restarts

Lemma 4.1 (Estimator decomposition and noise control)**.** *Under bounded rewards, twice differentiable policies with bounded score and Hessian on $\mathcal{K}$, and finite second moments of the likelihood-ratio estimators, the finite-unroll Meta-MAPG estimator satisfies Assumption 2. In particular, the current-policy, own-learning, and peer-learning components can each be written as a conditional mean plus a martingale-difference error; finite unroll and opponent-modeling truncation contribute the bias b_n.*

Lemma 4.2 (Local Meta-MAPG convergence)**.** *Suppose Assumptions 1–5 hold and no restarts occur. If ϕ_0 lies in the attracting neighbourhood of an SOS Nash equilibrium $\phi^* \in \mathcal{N}$, then*

$$\phi_n \to \phi^* \qquad \textit{almost surely.}$$

Moreover, under the quantitative bias and variance exponents of Giannou et al. (2022), the usual local squared-distance rate is inherited.

Proposition 4.3 (Local drift improvement at the shifted fixed point)**.** *Fix an SOS Nash equilibrium ϕ^* of v with drift constant $\mu > 0$ and $v, M \in C^1$ near ϕ^*. The own-learning component vanishes at the Nash, $M^{\text{own}}(\phi^*) = 0$, because $\nabla_{\phi_i} J_i(\phi^*) = 0$. The peer-learning component satisfies $M_i^{\text{peer}}(\phi^*) = \eta[\nabla_{\phi_i}\nabla_{\phi_j} J_j(\phi^*)]^\top \nabla_{\phi_j} J_i(\phi^*)$, which need not vanish in general-sum games because $\nabla_{\phi_j} J_i(\phi^*) \neq 0$ in general. Since $J_v := Dv(\phi^*)$ is invertible under SOS, the implicit function theorem gives a C^1 branch $\lambda \mapsto \phi^*_\lambda$ with $F_\lambda(\phi^*_\lambda) = 0$ and $\|\phi^*_\lambda - \phi^*\| = O(\lambda)$. Write $J_M := DM(\phi^*)$ and*

$$\mu_M := -\lambda_{\max}\left(\tfrac{J_M + J_M^\top}{2}\right).$$

Weyl's inequality gives

$$-\lambda_{\max}\left(\tfrac{J_\lambda + J_\lambda^\top}{2}\right) \ge \mu + \lambda\mu_M - O(\lambda^2), \quad J_\lambda := DF_\lambda(\phi^*_\lambda). \tag{2}$$

If DF_λ is L_λ-Lipschitz on $\mathcal{B}_{\rho_\lambda}(\phi^*_\lambda)$, then on that ball with $\rho_\lambda := (\mu + \lambda\mu_M)/(2L_\lambda)$ (up to $O(\lambda^2)$),

$$\langle F_\lambda(\phi), \phi - \phi^*_\lambda \rangle \le -\tfrac{1}{2}(\mu + \lambda\mu_M)\|\phi - \phi^*_\lambda\|^2.$$

The certified SOS radius around the shifted *fixed point ϕ^*_λ scales at least linearly in λ whenever $\mu_M > 0$. Under the annealed schedule (Theorem 4.4) the shift is transient, $\phi^*_\lambda \to \phi^*$ as $\lambda_n \to 0$, so the asymptotic target remains the base-game Nash. If $\mu_M \le 0$, opponent-aware shaping does not enlarge the certified local basin.*

Theorem 4.4 (Two-phase Meta-MAPG)**.** *Fix an SOS Nash equilibrium ϕ^* and, for small $\lambda_c > 0$, let $\phi^*_{\lambda_c}$ be the shifted fixed point of $F_{\lambda_c} = v + \lambda_c M$ from Theorem 4.3. Let $\mathcal{B}_\rho = \{\phi : \|\phi - \phi^*_{\lambda_c}\| \le \rho\}$ be the phase-1 SOS ball from Theorem 4.3 with drift constant $\mu_{\lambda_c} > 0$; by $\|\phi^*_{\lambda_c} - \phi^*\| = O(\lambda_c)$, it contains ϕ^* in its interior for small λ_c. Run Meta-MAPG in two phases. In phase 1, use constant shaping $\lambda_n = \lambda_c$ and constant step size α_0 for N_0 iterations; if*

$\phi_{N_0} \notin \mathcal{B}_{\rho/2}$, declare the epoch failed. In phase 2, started only from $\mathcal{B}_{\rho/2}$, use $\alpha_n = \alpha_0 n^{-p}$ and $\lambda_n = \lambda_c n^{-q}$ with $1/2 < p \le 1$ and $q > 1 - p$, together with Assumption 3.

If the phase-1 estimator has conditional variance at most σ^2 and bias norm at most β_0, then there is a constant C depending on local bounds of F_{λ_c} such that

$$\Pr(\phi_{N_0} \notin \mathcal{B}_{\rho/2} \mid \phi_0 \in \mathcal{B}_\rho) \le \frac{4}{\rho^2}\left(V_0 e^{-\mu_{\lambda_c}\alpha_0 N_0} + C\alpha_0(\beta_0^2 + \alpha_0\sigma^2)\right),$$

where $V_0 = \|\phi_0 - \phi^*_{\lambda_c}\|^2$. Hence for any target failure probability δ, choosing $N_0 = O((\mu_{\lambda_c}\alpha_0)^{-1}\log(V_0/(\delta\rho^2)))$ and α_0 small enough gives phase-2 entry with probability at least $1-\delta$. Conditional on this entry event, phase 2's annealed shaping pulls the fixed point back to ϕ^* and Theorem 4.2 gives $\phi_n \to \phi^*$ almost surely.

Corollary 4.5 (Basin-entry speedup at a common radius)**.** *Fix $\rho_\star := \min(\rho_0, \rho_\lambda) \le \rho_\lambda$, the largest ball on which* both *v and F_λ satisfy SOS drift. Under the drift-improvement condition of Theorem 4.3, the phase-1 horizon to reach $\mathcal{B}_{\rho_\star/2}$ scales as $N_0(\lambda) = O(1/(\alpha_0\mu_\lambda))$ with a constant matched across λ (same target radius $\rho_\star$, same δ). Thus, relative to PG, Meta-MAPG shaping reduces the certified basin-entry horizon by*

$$\frac{N_0(0)}{N_0(\lambda)} \gtrsim 1 + \frac{\lambda\mu_M}{\mu}$$

up to higher-order local-linearisation terms. If $\mu_M \le 0$, opponent-aware shaping need not enlarge the basin.

Theorem 4.6 (Restart globalisation of two-phase Meta-MAPG)**.** *Let each restart epoch run the two-phase procedure of Theorem 4.4 with the same phase-1 ball $\mathcal{B}_\rho = \{\phi : \|\phi - \phi^*_{\lambda_c}\| \le \rho\}$ on which F_{λ_c} satisfies the SOS drift condition with constant $\mu_{\lambda_c} > 0$. Let the restart distribution ν satisfy $p_\rho := \nu(\mathcal{B}_\rho) > 0$. If epochs are independent, then the process enters a successful phase-2 epoch after finitely many restarts almost surely. If T is the index of the first successful epoch, then*

$$\Pr(T > k) \le (1 - p_\rho(1-\delta))^k, \qquad \mathbb{E}[T] \le \frac{1}{p_\rho(1-\delta)}.$$

Thus, after finitely many restarts, the process enters an epoch where the local theorem applies and then converges almost surely to a stable Nash equilibrium in $\mathcal{N}$. This globalises a local theorem by search; it does not make the vector field globally contracting. In the worst case, if ν is roughly uniform on a radius-R d-dimensional compact set and $\mathcal{B}_\rho$ contains a radius-ρ ball, the bound scales as $O((R/\rho)^d)$. High-dimensional domains therefore need structured or warm-start restarts.

Remark 4.7 (Worst case vs empirics). The $(R/\rho)^d$ bound is worst-case for uniform ν; the empirical $\mathbb{E}[T] \approx 1.24$ in Stag Hunt (Section 7) reflects a non-uniform, near-warm-start restart distribution that concentrates mass in the cooperative basin rather than any tightness of the bound.

5. Why the Theory Works

Step 1: estimator decomposition. This is the Meta-MAPG-specific step. The theorem of Kim et al. (2021) splits the sampled gradient into current, own-learning, and peer-learning score-function terms. Bounded rewards and scores give finite conditional variance; subtracting the conditional mean gives martingale noise. Finite unrolls and approximate opponent models give the bias b_n, controlled by Assumption 3.

Step 2: summability. The bias and variance terms are checked against the stochastic-approximation conditions. Because $\sum_n \alpha_n^2 < \infty$, the square-integrable martingale noise is summable in the Robbins–Siegmund sense (Robbins & Monro, 1951; Robbins & Siegmund, 1971; Hall & Heyde, 1980). Because $\sum_n \alpha_n\beta_n < \infty$, the deterministic truncation bias is also summable.

Step 3: local theorem transfer. Map the Meta-MAPG update to Giannou et al.'s local PG template. In the annealed regime, the limiting field is Nash-consistent: local stationary points are the same stable Nash policies as for the base game-gradient field. The Lyapunov recursion is the same as in Giannou et al. (2022), up to summable forcing terms.

Step 4: two-phase transfer. Theorem 4.4 separates finite-time shaping from asymptotic convergence. Phase 1 uses constant shaping only for basin entry, with drift constant μ_{λ_c} from Theorem 4.3. After entry into $\mathcal{B}_{\rho/2}$, the schedule anneals; Theorem 4.2 follows from the local attraction theorem for stable SOS Nash policies. No global monotonicity is required. The argument is local, as in Giannou et al.

Step 5: restart globalisation. Restarts globalise the local result by a geometric argument. Each epoch is independent. The restart distribution assigns mass p_ρ to the phase-1 region, and the two-phase theorem reaches phase 2 with probability at least $1-\delta$. Each epoch therefore succeeds with probability at least $p_\rho(1-\delta)$. The first successful epoch is geometrically dominated. Basin maps in Section 7 show how opponent-aware shaping can improve this constant by increasing basin-entry probability.

6. Related Work

Positioning. Prior work builds opponent-shaping algorithms. LOLA differentiates through an opponent update (Foerster et al., 2018a); SOS stabilises shaping in differentiable games (Letcher et al., 2019); COLA enforces consis-

tency of mutual shaping (Willi et al., 2022); POLA adds a proximal view (Zhao et al., 2022); M-FOS learns long-horizon shaping policies without a differentiable opponent model (Lu et al., 2022). Our paper sits closest to Kim et al. (2021): we inherit the current/own/peer decomposition and ask a narrower question, namely when finite-unroll Meta-MAPG fits stochastic approximation, improves basin entry, and preserves the asymptotic Nash target.

Kim et al. study Meta-MAPG as a general meta-learning algorithm for rapid adaptation in mixed-incentive, competitive, and cooperative environments. Our focus is narrower and more local: we isolate the stochastic-approximation structure needed for convergence, make the basin-shaping role of the peer term explicit, and use restarts as the mechanism that turns a local theorem into an equilibrium-search statement.

7. Experiments

The experiments are small by design. We use tabular Bernoulli policies in Stag Hunt and finite-horizon iterated Prisoner's Dilemma (IPD). Stag Hunt is the core test: two stable Nash equilibria, non-trivial basins, and a two-dimensional policy space where the drift constants μ and μ_M in Theorem 4.3 can be measured. It is also the standard testbed for LOLA, SOS, and COLA (Foerster et al., 2018a; Letcher et al., 2019; Willi et al., 2022). IPD tests the same estimator in a finite-horizon sequential game. Our Stag Hunt is a one-state, one-step, two-agent Markov game. Each agent samples C or D from a Bernoulli policy, with row-player payoff matrix

	C	D
C	4	0
D	3	2

and the column-player payoff given by the transpose. Thus (C, C) is payoff-dominant with social welfare 8, while (D, D) is risk-dominant with social welfare 4. In the figures, each coordinate is an agent's cooperation probability $p_i = \pi_i(C)$. Success means both final cooperation probabilities exceed 0.82.

Updates are sample-based: for a trajectory score $s_i = \nabla_{\phi_i} \log p_\phi(\tau)$ and return R_i, the base estimator is $\hat{g}_i = \frac{1}{B}\sum_b R_i^{(b)} s_i^{(b)}$. The own-learning term uses the likelihood-ratio Hessian estimator

$$\widehat{H}_{ii} = \frac{1}{B}\sum_b R_i^{(b)} \left(s_i^{(b)} s_i^{(b)\top} + \nabla^2_{\phi_i} \log p_\phi(\tau^{(b)}) \right),$$

and the peer-learning term uses the cross-score estimator

$$\widehat{C}_{ji} = \frac{1}{B}\sum_b R_j^{(b)} s_j^{(b)} s_i^{(b)\top}.$$

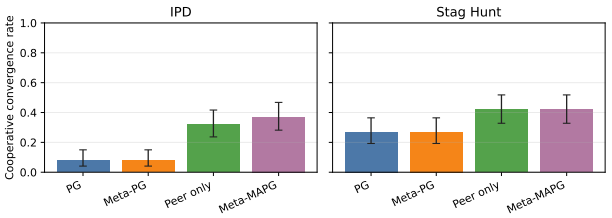

Figure 1. Component ablation with Wilson 95% confidence intervals over 100 seeds. Peer-aware methods (peer-only, Meta-MAPG) separate from non-peer methods (PG, Meta-PG) in both games. Peer-only and full Meta-MAPG are not distinguishable within CIs; the claim is the peer-learning effect, not strict dominance of the full method.

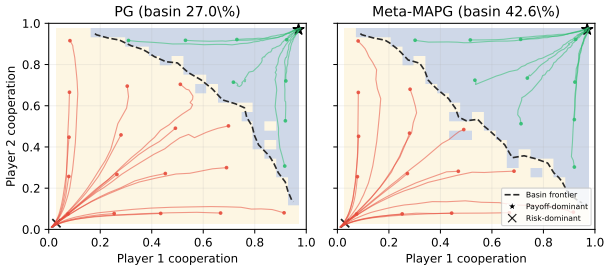

Figure 2. Basin geometry and dynamics in Stag Hunt. Background shading is the empirical basin map on a 21×21 grid (lavender = payoff-dominant, cream = risk-dominant). The dashed contour is the empirical separatrix. Overlaid trajectories start from a 5×5 sub-grid; green curves succeed, red curves fail. Meta-MAPG pushes the frontier left and down: the cooperative basin grows from 27.0% under PG to 42.6%, consistent with Theorem 4.3.

The implemented one-step correction is $\eta \widehat{H}_{ii}^\top \hat{g}_i$ for own learning and $\eta \widehat{C}_{ji}^\top \widehat{\nabla_{\phi_j} J_i}$ for peer learning. We compare PG, Meta-PG (no peer term), peer-only LOLA-style shaping (no own term), and full Meta-MAPG. Unless stated otherwise, summaries use 100 seeds; restart summaries pair initial policies across all four methods; restart-budget curves pair PG and Meta-MAPG starts; all main experiments use the same peer coefficient. Raw CSVs and run configuration are included.

The ablation shows that peer-learning terms matter; it does not show that full Meta-MAPG dominates every LOLA-style variant. Peer-only and full Meta-MAPG are indistinguishable within CIs in both Stag Hunt (42% vs 42%, 1.24 vs 1.24 restarts) and IPD (32% vs 37% within Wilson CI), so the empirical opponent-aware signal reduces to the peer term; disentangling the own-term likely requires longer unrolls (an $L \in \{1, 3, 5\}$ sweep in IPD is a natural follow-up). The trajectory figure shows the finite-time mechanism: opponent-aware shaping changes the local flow where the basin map changes. The restart-budget, basin, and peer-sweep experiments support Theorems 4.3 to 4.6: restarts turn local convergence into equilibrium search, and shaping

Table 1. Positioning among opponent-aware methods. "Nash-preserving" means the stated theory preserves the original game's stable Nash target rather than merely optimising a shaped meta-objective.

Method	Convergence guarantee	Shaping mechanism	Opponent model	Nash-preserving
LOLA	No general	1-step gradient	White-box	No
SOS	Differentiable games	Stable OS	White-box	SFP-focused
COLA	Simple-game stability	Consistent OS	White-box	No
POLA	Empirical/proximal	Proximal OS	Learned	Not primary
M-FOS	Empirical	Meta-policy OS	No	No
This work	Local SA + restarts	Two-phase Meta-MAPG	Sample estimates	Yes, by annealing

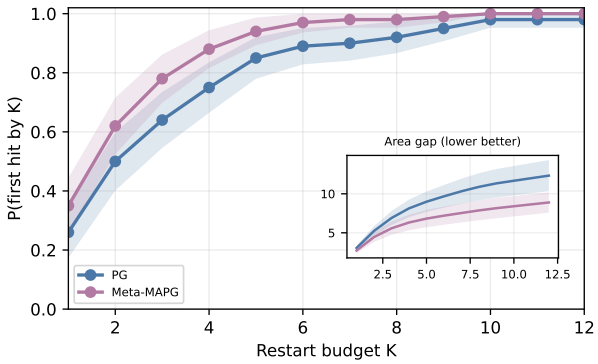

Figure 3. Restart budget as equilibrium-selection budget in Stag Hunt. With paired restart initialisations, we plot the probability of discovering the payoff-dominant equilibrium by budget K; bands show 95% confidence intervals. The inset shows $\sum_{k \le K}(8 - W_k)$, the cumulative gap below best welfare (point estimates only, no CI band). Lower is better. Meta-MAPG reaches the good basin with fewer restarts.

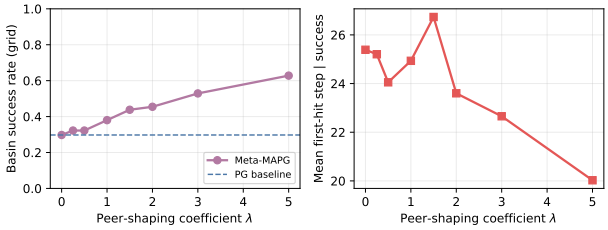

Figure 4. Peer-shaping sweep on Stag Hunt. Left: payoff-dominant basin fraction on an 11×11 initial-policy grid as peer coefficient λ varies. The rate grows from the PG baseline (29.8%) to 62.8% at $\lambda = 5$, roughly linearly for small λ, as predicted by $\rho_\lambda \gtrsim \rho_0 + \lambda\mu_M/(2L)$. Right: conditional mean first-hit step contracts from ≈ 25 to ≈ 20, matching Theorem 4.5.

Table 2. Main 100-seed summary. Cooperative success uses a minimum initial-state cooperation threshold of 0.82. Restart success and mean restarts are reported for all methods using paired restart initialisations.

Game	Method	Ablation success	Restart success	Restarts
stag hunt	PG	27%	100%	1.71
stag hunt	Meta-PG	27%	100%	1.75
stag hunt	Peer only	42%	100%	1.24
stag hunt	Meta-MAPG	42%	100%	1.24
ipd	PG	8%	51%	8.68
ipd	Meta-PG	8%	57%	7.90
ipd	Peer only	32%	97%	2.91
ipd	Meta-MAPG	37%	100%	2.13

improves that search by increasing cooperative basin mass and first-passage probability. Figure 5 runs the two-phase schedule directly. It shows that annealing after N_0 steps preserves basin-entry performance while restoring the Nash target. The MLP-policy IPD check in Section G is a transfer check, not a deep-MARL benchmark. The estimator sanity check in Section H is not a proof of Assumption 2; it checks that the sample estimator moves in the predicted direction as batch size grows.

8. Discussion

The theorem is local first. Meta-MAPG inherits Giannou et al.'s stable-Nash theory only inside attracting neighbourhoods and only after the estimator has stochastic-approximation form. The two-phase theorem fixes the algorithmic role: constant shaping is for basin entry; annealing restores the original Nash target. Restarts search across basins. Selecting the best endpoint makes restart budget an equilibrium-selection resource. Its cost depends on basin probability p_ρ; shaping helps only when it increases that probability.

The assumptions are real. The estimator must have bounded conditional variance. Truncation must make unroll bias summable. Asymptotic shaping must be Nash-consistent. Constant shaping is useful for basin geometry, but a constant peer term can move fixed points in general-sum games. This is why the theorem separates annealed convergence from constant-shaping diagnostics.

The experiments are deliberately limited. Matrix and tabular games let us measure the theorem's objects: basins, restarts, drift, and estimator variance. Naive random restarts degrade with dimension: a radius-ρ basin inside a radius-R search region has worst-case mass $(\rho/R)^d$. Deep MARL needs stronger variance control, opponent models, differentiable longer unrolls, and structured restarts such as warm starts

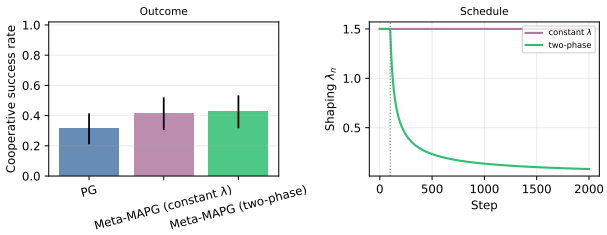

Figure 5. Two-phase schedule vs constant shaping in Stag Hunt (80 seeds, 2000 steps, $\lambda_c = 1.5$). PG reaches 31.25%, constant-shaping Meta-MAPG reaches 41.25%, and the two-phase schedule reaches 42.50%; second-half cooperation std is $\leq 5 \times 10^{-4}$ in all three, indicating no endpoint drift at this horizon. In tabular Stag Hunt the constant-shaping fixed-point shift is empirically small, so Nash-consistency under annealing (Theorem 2.2 and Assumption 4) is not stressed here; non-tabular parameterisations would stress-test it. Right: λ_n over time; the dotted line marks $N_0 = 100$.

around high-welfare policies, possibly with DiCE-style estimators (Foerster et al., 2018b). The claim here is narrower: opponent-aware gradients can fit stochastic approximation, and basin-entry efficiency is the right global metric when convergence is local.

We do not claim unconditional convergence of arbitrary opponent-aware learning in deep MARL. We show that finite-unroll Meta-MAPG fits classical stochastic approximation under boundedness, smoothness, Nash-consistency, and summability assumptions. Independent restarts globalise the local result when local basin mass is positive. The MLP-policy check in Section G only tests whether the peer-vs-nonpeer separation survives a modest neural parameterisation.

A. Proof of Theorem 4.1

For a sampled trajectory τ, write $S_i(\tau) = \nabla_{\phi_i} \log p_\phi(\tau)$, $H_i(\tau) = \nabla^2_{\phi_i} \log p_\phi(\tau)$, and let $R_i(\tau)$ be agent i's discounted return.

Current term. The likelihood-ratio identity (Williams, 1992; Sutton & Barto, 2018) gives the single-sample estimator $\widehat{G}_i^{\mathrm{cur}}(\tau) := R_i(\tau)S_i(\tau)$ with $\mathbb{E}[\widehat{G}_i^{\mathrm{cur}} \mid \mathcal{F}_n] = \nabla_{\phi_i} J_i(\phi_n)$.

Own-learning term. Differentiating $J_i(\phi_i + \eta\nabla_{\phi_i} J_i, \phi_{-i})$ in ϕ_i yields $\nabla_{\phi_i} J_i + \eta[\nabla^2_{\phi_i} J_i]^\top \nabla_{\phi_i} J_i$, and the Hessian identity

$$\nabla^2_{\phi_i} J_i = \mathbb{E}\big[R_i(\tau)\big(S_i(\tau)S_i(\tau)^\top + H_i(\tau)\big)\big]$$

gives the single-sample estimator $\widehat{G}_i^{\mathrm{own}}(\tau) := \eta\, \widehat{H}_{ii}(\tau)^\top \widehat{G}_i^{\mathrm{cur}}(\tau)$.

Peer-learning term. For $\phi'_j = \phi_j + \eta\nabla_{\phi_j} J_j$, differentiating $J_i(\phi_i, \phi'_j)$ in ϕ_i contributes $\eta[\nabla_{\phi_i}\nabla_{\phi_j} J_j]^\top \nabla_{\phi_j} J_i$; under per-agent parameterisation S_j has no ϕ_i-dependence, so the mixed likelihood-ratio identity collapses to

$$\nabla_{\phi_i}\nabla_{\phi_j} J_j = \mathbb{E}\big[R_j(\tau)S_j(\tau)S_i(\tau)^\top\big],$$

yielding $\widehat{G}_i^{\mathrm{peer}}(\tau) := \eta\, \widehat{C}_{ji}(\tau)^\top \widehat{G}_j^{\mathrm{cur}}(\tau)$.

Martingale noise and truncation bias. Each of $\widehat{G}_i^{\mathrm{cur}}, \widehat{G}_i^{\mathrm{own}}, \widehat{G}_i^{\mathrm{peer}}$ is a polynomial in the bounded-second-moment quantities R_i, R_j, S_i, S_j, H_i, hence has finite conditional second moment on $\mathcal{K}$. Setting

$$\xi^\bullet_{n+1} := \widehat{G}_i^\bullet - \mathbb{E}[\widehat{G}_i^\bullet \mid \mathcal{F}_n], \quad \bullet \in \{\mathrm{cur}, \mathrm{own}, \mathrm{peer}\},$$

each is a martingale difference with bounded conditional variance; so is their sum ξ_{n+1}. Longer unrolls repeat this chain-rule structure across L_n inner updates, and the truncated-tail plus approximate-opponent modelling residual is deterministic given $\mathcal{F}_n$. Collecting it into $b_n := \mathbb{E}[g_n \mid \mathcal{F}_n] - F(\phi_n)$, the bounded-derivative hypotheses on $\mathcal{K}$ and Assumption 3 give $\|b_n\| \leq C\beta_n$ with $\beta_n \to 0$. Assembling the three components produces

$$g_n = F(\phi_n) + b_n + \xi_{n+1},$$

which is exactly Assumption 2. □

B. Local Lyapunov Drift

We spell out the stochastic-approximation step behind Theorem 4.2. The goal is not to reprove Giannou et al. (2022); it is to show that Meta-MAPG has the same Lyapunov form after decomposition.

Lemma B.1 (One-step local Lyapunov drift)**.** *Fix an SOS Nash equilibrium ϕ^* and a ball $\mathcal{B}_\rho = \{\phi : \|\phi - \phi^*\| \leq \rho\}$ on which the effective field F satisfies*

$$\langle F(\phi), \phi - \phi^* \rangle \leq -\mu\|\phi - \phi^*\|^2.$$

Suppose the stopped iterates remain in $\mathcal{B}_\rho$ and the Meta-MAPG update admits

$$g_n = F(\phi_n) + \xi_{n+1} + b_n,$$
$$\mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] = 0, \qquad \mathbb{E}[\|\xi_{n+1}\|^2 \mid \mathcal{F}_n] \leq \sigma^2,$$

with $\|b_n\| \leq \beta_n$. If $V_n = \|\phi_n - \phi^*\|^2$ and $\|F(\phi)\| \leq C_F$ on $\mathcal{B}_\rho$, then

$$\mathbb{E}[V_{n+1} \mid \mathcal{F}_n] \leq (1 - 2\mu\alpha_n)V_n + 2\alpha_n\beta_n\sqrt{V_n} + 3\alpha_n^2(C_F^2 + \beta_n^2 + \sigma^2). \tag{3}$$

Proof. Projection onto the convex compact set $\mathcal{K}$ is nonexpansive, so it is enough to bound the unprojected step.

Expanding the square gives

$$\begin{aligned} V_{n+1} &\le \|\phi_n - \phi^* + \alpha_n g_n\|^2 \\ &= V_n + 2\alpha_n \langle g_n, \phi_n - \phi^* \rangle + \alpha_n^2 \|g_n\|^2. \end{aligned}$$

Taking conditional expectation and using $\mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] = 0$,

$$\begin{aligned} \mathbb{E}[\langle g_n, \phi_n - \phi^* \rangle \mid \mathcal{F}_n] &= \langle F(\phi_n), \phi_n - \phi^* \rangle \\ &\quad + \langle b_n, \phi_n - \phi^* \rangle \\ &\le -\mu V_n + \beta_n \sqrt{V_n}. \end{aligned}$$

For the quadratic term, $(a+b+c)^2 \le 3(a^2+b^2+c^2)$ gives

$$\begin{aligned} \mathbb{E}[\|g_n\|^2 \mid \mathcal{F}_n] &\le 3\big(\|F(\phi_n)\|^2 + \|b_n\|^2 \\ &\quad + \mathbb{E}[\|\xi_{n+1}\|^2 \mid \mathcal{F}_n]\big) \\ &\le 3(C_F^2 + \beta_n^2 + \sigma^2). \end{aligned}$$

Substituting the last two displays into the square expansion proves (3). □

If $\sum_n \alpha_n \beta_n < \infty$ and $\sum_n \alpha_n^2 < \infty$, then (3) is in Robbins–Siegmund form. Indeed,

$$2\alpha_n \beta_n \sqrt{V_n} \le \alpha_n \beta_n (1 + V_n),$$

so for all large n,

$$\mathbb{E}[V_{n+1} \mid \mathcal{F}_n] \le (1 - a_n) V_n + \eta_n$$

with $\sum_n a_n = \infty$ and $\sum_n \eta_n < \infty$. Robbins–Siegmund yields $V_n \to V_\infty$ almost surely and $\sum_n a_n V_n < \infty$; since $\sum_n a_n = \infty$, $V_\infty = 0$. Thus $\phi_n \to \phi^*$ almost surely on the local event.

C. Proof of Proposition 4.3

We prove the three parts separately.

Fixed-point shift. $F_\lambda(\phi^*) = \lambda M(\phi^*)$, where $M(\phi^*) = M^{\text{peer}}(\phi^*) \neq 0$ in general (own-learning vanishes because $\nabla_{\phi_i} J_i(\phi^*) = 0$). Since $J_v = Dv(\phi^*)$ is non-singular under SOS, the implicit function theorem applied to $\Phi(\lambda, \phi) := F_\lambda(\phi)$ at $(0, \phi^*)$ gives a C^1 branch ϕ_λ^* with $F_\lambda(\phi_\lambda^*) = 0$ and

$$\phi_\lambda^* = \phi^* - \lambda J_v^{-1} M(\phi^*) + O(\lambda^2),$$

so $\|\phi_\lambda^* - \phi^*\| = O(\lambda)$ for small λ.

Linearised drift bound. Both J_v and J_M are real matrices; their symmetric parts $\mathrm{sym}(A) := (A + A^\top)/2$ are Hermitian. By definition $-\lambda_{\max}(\mathrm{sym}(J_v)) = \mu$ and $-\lambda_{\max}(\mathrm{sym}(J_M)) = \mu_M$. At the shifted fixed point, $DF_\lambda(\phi_\lambda^*) = J_v + \lambda J_M + O(\lambda^2)$ by C^1 dependence on ϕ and the $O(\lambda)$ shift. Weyl's inequality gives

$$\lambda_{\max}\big(\mathrm{sym}(J_v + \lambda J_M)\big) \le \lambda_{\max}\big(\mathrm{sym}(J_v)\big) + \lambda\, \lambda_{\max}\big(\mathrm{sym}(J_M)\big),$$

and negating, together with the $O(\lambda^2)$ evaluation shift, yields (2).

Certified radius. Let $\mu_\lambda := \mu + \lambda \mu_M$. Fix $\phi \in \mathcal{B}_{\rho_\lambda}(\phi_\lambda^*)$ and write $x := \phi - \phi_\lambda^*$. Since $F_\lambda(\phi_\lambda^*) = 0$, Taylor's theorem with integral remainder gives

$$F_\lambda(\phi) = J_\lambda x + R_\lambda(x), \qquad \|R_\lambda(x)\| \le \tfrac{1}{2} L_\lambda \|x\|^2,$$

with $J_\lambda = DF_\lambda(\phi_\lambda^*)$ and L_λ the Lipschitz constant of DF_λ on $\mathcal{B}_{\rho_\lambda}(\phi_\lambda^*)$. Taking inner products,

$$\langle J_\lambda x, x \rangle = x^\top \mathrm{sym}(J_\lambda) x \le -\mu_\lambda \|x\|^2$$

by Rayleigh and (2) (absorbing the $O(\lambda^2)$ term into μ_λ for small λ), and $|\langle R_\lambda(x), x \rangle| \le \frac{1}{2} L_\lambda \|x\|^3$. Hence

$$\langle F_\lambda(\phi), \phi - \phi_\lambda^* \rangle \le -\mu_\lambda \|x\|^2 + \tfrac{1}{2} L_\lambda \|x\|^3.$$

For $\|x\| \le \rho_\lambda := \mu_\lambda / (2L_\lambda)$, the cubic term is at most $(\mu_\lambda/4)\|x\|^2$, so

$$\langle F_\lambda(\phi), \phi - \phi_\lambda^* \rangle \le -\tfrac{1}{2} \mu_\lambda \|x\|^2,$$

the stated SOS drift on $\mathcal{B}_{\rho_\lambda}(\phi_\lambda^*)$. □

D. Proof of Theorem 4.4

Write $V_n = \|\phi_n - \phi_{\lambda_c}^*\|^2$ (phase-1 Lyapunov, centred at the shifted fixed point of Theorem 4.3) and let $\tau := \inf\{n \le N_0 : \phi_n \notin \mathcal{B}_\rho\}$ with $\tau = N_0 + 1$ if the iterates never leave. On the event $\{n < \tau\}$ the phase-1 drift of Theorem 4.3 applies with constant μ_{λ_c}.

Step 1: one-step drift. On $\{n < \tau\}$, by Assumption 2 and the phase-1 drift bound around $\phi_{\lambda_c}^*$,

$$\begin{aligned} \mathbb{E}[V_{n+1} \mid \mathcal{F}_n] &= \mathbb{E}\big[\|\phi_n - \phi_{\lambda_c}^* + \alpha_0 g_n\|^2 \mid \mathcal{F}_n\big] \\ &= V_n + 2\alpha_0 \langle F_{\lambda_c}(\phi_n), \phi_n - \phi_{\lambda_c}^* \rangle \\ &\quad + 2\alpha_0 \langle b_n, \phi_n - \phi_{\lambda_c}^* \rangle + \alpha_0^2 \mathbb{E}[\|g_n\|^2 \mid \mathcal{F}_n] \\ &\le V_n - 2\alpha_0 \mu_{\lambda_c} V_n + 2\alpha_0 \beta_0 \sqrt{V_n} + \alpha_0^2 G^2, \end{aligned}$$

where $G^2 := 3(C_{F_{\lambda_c}}^2 + \beta_0^2 + \sigma^2)$ is a uniform bound on the conditional second moment of g_n (with $C_{F_{\lambda_c}}$ the sup-norm of F_{λ_c} on $\mathcal{B}_\rho$). Applying $2ab \le \eta a^2 + \eta^{-1} b^2$ with $a = \sqrt{V_n}$, $b = \beta_0$, and $\eta = \mu_{\lambda_c}$,

$$\mathbb{E}[V_{n+1} \mid \mathcal{F}_n] \le (1 - \mu_{\lambda_c} \alpha_0) V_n + \frac{\alpha_0 \beta_0^2}{\mu_{\lambda_c}} + \alpha_0^2 G^2.$$

Step 2: unconditional recursion on the stopped process. Define the stopped sequence $\tilde{V}_n := V_{n \wedge \tau}$. The drift bound and the fact that $\tilde{V}_{n+1} = \tilde{V}_n$ on $\{n \ge \tau\}$ give

$$\mathbb{E}[\tilde{V}_{n+1}] \le (1 - \mu_{\lambda_c} \alpha_0) \mathbb{E}[\tilde{V}_n] + c_1,$$

$$c_1 := \frac{\alpha_0 \beta_0^2}{\mu_{\lambda_c}} + \alpha_0^2 G^2.$$

Iterating,

$$\mathbb{E}[\tilde{V}_{N_0}] \le (1-\mu_{\lambda_c}\alpha_0)^{N_0} V_0 + \frac{c_1}{\mu_{\lambda_c}\alpha_0}.$$

Using $(1-x)^k \le e^{-xk}$ for $x \in [0,1]$,

$$\mathbb{E}[\tilde{V}_{N_0}] \le V_0\, e^{-\mu_{\lambda_c}\alpha_0 N_0} + \frac{\beta_0^2}{\mu_{\lambda_c}^2} + \frac{\alpha_0 G^2}{\mu_{\lambda_c}}.$$

Step 3: Markov's inequality. If $\phi_{N_0} \notin \mathcal{B}_{\rho/2}$ and $\tau > N_0$, then $V_{N_0} \ge (\rho/2)^2$. If instead $\tau \le N_0$, then by definition of τ we have $\phi_\tau \notin \mathcal{B}_\rho$, so $V_\tau > \rho^2 \ge (\rho/2)^2$, and the stopped process freezes at $\tilde{V}_{N_0} = V_\tau$. Hence in both cases $\tilde{V}_{N_0} \ge (\rho/2)^2$. Therefore

$$\begin{aligned}&\Pr(\phi_{N_0} \notin \mathcal{B}_{\rho/2} \mid \phi_0 \in \mathcal{B}_\rho)\\ &\quad \le \Pr(\tilde{V}_{N_0} \ge (\rho/2)^2) \le \frac{4\mathbb{E}[\tilde{V}_{N_0}]}{\rho^2}.\end{aligned}$$

Combining with Step 2,

$$\begin{aligned}&\Pr(\phi_{N_0} \notin \mathcal{B}_{\rho/2} \mid \phi_0 \in \mathcal{B}_\rho)\\ &\quad \le \frac{4}{\rho^2}\left(V_0 e^{-\mu_{\lambda_c}\alpha_0 N_0} + C\alpha_0(\beta_0^2 + \alpha_0\sigma^2)\right),\end{aligned}$$

for a constant C depending on $\mu_{\lambda_c}, C_{F_{\lambda_c}}$.

Step 4: choice of N_0. Given a target $\delta \in (0,1)$, set

$$N_0 = \left\lceil \frac{1}{\mu_{\lambda_c}\alpha_0} \log\frac{8V_0}{\delta\rho^2} \right\rceil, \qquad \alpha_0 \le \frac{\delta\rho^2}{8C(\beta_0^2+\sigma^2)}.$$

Each summand in Step 3 is then at most $\delta/2$, so the total failure probability is at most δ.

Step 5: phase 2. Conditional on $\phi_{N_0} \in \mathcal{B}_{\rho/2}$, the subsequent recursion uses $\alpha_n = \alpha_0 n^{-p}$ and $\lambda_n = \lambda_c n^{-q}$ with $1/2 < p \le 1$ and $q > 1-p$, so $\sum_n \alpha_n = \infty$, $\sum_n \alpha_n^2 < \infty$, and $\sum_n \alpha_n\lambda_n < \infty$ by direct integral comparison. The annealed fixed-point branch $\phi^*_{\lambda_n} \to \phi^*$ by Theorem 4.3, and since $\phi^* \in \mathcal{B}_{\rho/2} \subset \mathcal{B}_\rho$ by the small-λ_c hypothesis of Theorem 4.4, together with Assumptions 2, 3 and 5 the hypotheses of Theorem 4.2 are satisfied on the event $\{\phi_{N_0} \in \mathcal{B}_{\rho/2}\}$. Conditional on this event, $\phi_n \to \phi^*$ almost surely. The total probability that the two-phase procedure converges in a single epoch is therefore at least $(1-\delta)\cdot 1 = 1-\delta$. □

E. Proof of Theorem 4.6

Let $\mathcal{F}_k^{\text{epoch}}$ denote the σ-algebra generated by epochs $1,\dots,k-1$ (outcomes only; restart draws are independent by assumption). Let A_k be the event that epoch k starts in $\mathcal{B}_\rho$ and that the two-phase procedure of Theorem 4.4 reaches phase 2 and converges within that epoch.

Step 1: per-epoch lower bound. Restart k samples $\phi_0^{(k)} \sim \nu$ independently of $\mathcal{F}_k^{\text{epoch}}$, so

$$\Pr(\phi_0^{(k)} \in \mathcal{B}_\rho \mid \mathcal{F}_k^{\text{epoch}}) = \nu(\mathcal{B}_\rho) = p_\rho.$$

Conditional on $\phi_0^{(k)} \in \mathcal{B}_\rho$, Theorem 4.4 gives $\Pr(\text{epoch succeeds}) \ge 1-\delta$. Therefore

$$\Pr(A_k \mid \mathcal{F}_k^{\text{epoch}}) \ge p_\rho(1-\delta) =: p_\star > 0.$$

Step 2: geometric tail. Let $T := \inf\{k \ge 1 : A_k \text{ occurs}\}$. Then

$$\begin{aligned}\Pr(T > k) &= \mathbb{E}\left[\prod_{j=1}^{k} \mathbf{1}_{A_j^c}\right]\\ &= \mathbb{E}\left[\mathbb{E}[\mathbf{1}_{A_k^c} \mid \mathcal{F}_k^{\text{epoch}}] \prod_{j=1}^{k-1} \mathbf{1}_{A_j^c}\right].\end{aligned}$$

By Step 1, $\mathbb{E}[\mathbf{1}_{A_k^c} \mid \mathcal{F}_k^{\text{epoch}}] \le 1-p_\star$ almost surely, so iterating,

$$\Pr(T > k) \le (1-p_\star)^k.$$

Hence $\Pr(T < \infty) = 1$ and

$$\mathbb{E}[T] = \sum_{k\ge 0} \Pr(T > k) \le \sum_{k \ge 0} (1-p_\star)^k = \frac{1}{p_\star}.$$

Step 3: dimensional dependence. If ν is lower-bounded by density $m > 0$ on a region containing $\mathcal{B}_\rho(\phi^*)$ in $\mathbb{R}^d$, then $p_\rho \ge m\, c_d \rho^d$ where $c_d = \pi^{d/2}/\Gamma(d/2+1)$ is the volume of the unit d-ball. Combined with the certified radius $\rho_\lambda \ge (\mu + \lambda\mu_M)/(2L_\lambda)$ from Theorem 4.3, the expected number of epochs obeys

$$\mathbb{E}[T] \le \frac{1}{(1-\delta)mc_d}\left(\frac{2L_\lambda}{\mu+\lambda\mu_M}\right)^d.$$

Shaping reduces the base of this dimensional factor whenever $\mu_M > 0$; the overall worst-case dependence is still exponential in d, as discussed in Section 4. □

F. Experiment Reproducibility

The experiment scripts, figures, CSVs, and run configurations are included. The tabular script saves its command-line configuration and outputs under `artifacts/main`. These files cover the ablation, restart, trajectory, basin, peer-sweep, annealing, and estimator-sanity figures. Updates use no exact gradients; expected returns are computed only for logging and evaluation.

The optional MLP-policy IPD check in Section G requires PyTorch and uses DiCE-style surrogate losses for the higher-order terms. Its aggregate results are stored under `artifacts/mlp`. The repository README contains the full commands for regenerating both sets of experiments.

G. Non-Tabular IPD Check

To check that the ablation is not purely tabular, we repeat IPD with a two-layer MLP policy for each agent: a 5-dimensional one-hot previous-action input, two tanh hidden layers of width 16, and one Bernoulli cooperation logit. This scales the policy from 5 tabular logits to about 385 parameters per agent. We keep the tabular budget, batch size, seed schedule, peer coefficient, and own coefficient. Higher-order terms use PyTorch autograd with DiCE-style surrogate losses (Foerster et al., 2018b).

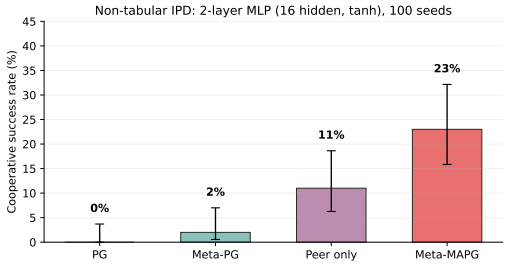

Figure 6. Non-tabular IPD check with MLP policies. Cooperative success rate over 100 seeds with Wilson 95% confidence intervals. Peer-aware methods remain above non-peer methods under the same coefficients as the tabular experiments: PG reaches 0%, Meta-PG 2%, peer-only 11%, and Meta-MAPG 23%. This is a transfer check, not a deep-MARL benchmark.

H. Estimator Sanity Check

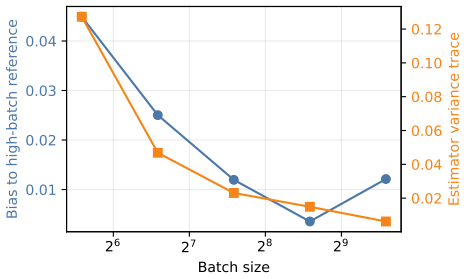

Figure 7. Estimator sanity check. At a fixed Stag Hunt parameter, the variance trace of the Meta-MAPG estimator decreases from 0.1272 at batch size 48 to 0.0062 at batch size 768. Bias is measured relative to a high-batch reference estimator, not an exact analytic gradient.

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